📚 Complete Tutorial Guide with Answers

Cybersecurity Game Tutorial Compendium
All Game Modes, Instructions, Questions & Answers

📖 Table of Contents

🟢 BEGINNER TUTORIALS

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1. Cybersecurity Fundamentals

📋 Overview

- **Tutorial ID:** `beginner_fundamentals`
- **Duration:** ~10 minutes
- **Format:** Multi-section learning + Quiz
- **Max Score:** 200 points (4 questions × 50 points each)
- **XP Rewards:**
  - 100% (200/200): 200 XP
  - 75%+ (150-199): 150 XP
  - 50-74% (100-149): 100 XP
  - Below 50%: 0 XP

📚 Learning Sections

Section 1: Introduction

What is Cybersecurity?

Cybersecurity = Protecting computers, networks, and data from attacks.

Core Topics:

- 🗄 The CIA Triad - 3 core principles of security
- 🎯 Threat Model - Understanding attackers

Section 2: CIA Triad - Confidentiality

Definition: Keeping secrets - only authorized people can access information.

Examples:

- ✓ Passwords (only you should know yours)
- ✓ Medical records (only you and your doctor)
- ✓ Credit card numbers (only you and the bank)
- ✓ Company secrets (trade secrets, financial data)

How it's protected:

- Encryption (scrambles data)
- Access controls (passwords, permissions)
- Authentication (proving who you are)

Breach Examples:

- ❌ Hacker steals customer database
- ❌ Employee leaks financial reports
- ❌ Someone reads your private messages

Section 3: CIA Triad - Integrity

Definition: Preventing tampering - data is accurate and unmodified.

Examples:

- ✓ Bank account balance (must be exact)
- ✓ Software downloads (no hidden malware)
- ✓ Medical prescriptions (correct dosage)
- ✓ Website content (not defaced)

How it's protected:

- Digital signatures (proves authenticity)
- Checksums/hashes (detects changes)
- Version control (tracks modifications)
- Access controls (limits who can edit)

Breach Examples:

- ❌ Hacker changes your bank balance
- ❌ Malware injected into software update
- ❌ Website modified to spread misinformation

Section 4: CIA Triad - Availability

Definition: Keeping services running - systems accessible when needed.

Examples:

- ✓ Website is online 24/7
- ✓ Email server responds quickly
- ✓ Hospital systems work during emergencies
- ✓ ATM machines dispense cash

How it's protected:

- Redundancy (backup servers)
- Load balancing (distribute traffic)
- DDoS protection (block attack traffic)
- Disaster recovery plans

Breach Examples:

- ❌ DDoS attack crashes website
- ❌ Ransomware locks all files
- ❌ Server outage (hardware failure)

Section 5: Threat Model

Understanding Threats, Vulnerabilities, and Risks

🎯 THREAT = Potential danger or attacker

- Examples: Hackers, malware, insider threats, natural disasters
- Think: "WHAT could attack us?"

🕳 VULNERABILITY = Weakness that can be exploited

- Examples: Outdated software, weak passwords, misconfiguration
- Think: "WHERE are we weak?"

⚠ RISK = Likelihood + Impact

- Formula: `Risk = Threat × Vulnerability × Impact`
- Think: "HOW bad could it be?"

Example:

```
Threat: Ransomware hackers
Vulnerability: No backups, outdated Windows
Risk: HIGH (if attacked, lose all files)

Mitigation: Update Windows + create backups
Result: Risk = LOW (can restore from backup)
```

✅ Quiz Questions & Answers

Question 1

Q: A hacker steals your password database. Which part of CIA Triad is violated?

Options:

- A) Confidentiality ✅ **CORRECT**
- B) Integrity
- C) Availability

Explanation: Confidentiality means keeping data SECRET. Stolen passwords = confidentiality breach!

Question 2

Q: An attacker modifies your website to show fake information. Which is violated?

Options:

- A) Confidentiality
- B) Integrity ✅ **CORRECT**
- C) Availability

Explanation: Integrity means data is ACCURATE and unmodified. Changed website = integrity breach!

Question 3

Q: A DDoS attack makes your website unreachable. Which is violated?

Options:

- A) Confidentiality
- B) Integrity
- C) Availability ✅ **CORRECT**

Explanation: Availability means services are ACCESSIBLE. Website down = availability breach!

Question 4

Q: What is the difference between Threat and Vulnerability?

Options:

- A) Threat = potential danger, Vulnerability = weakness ✅ **CORRECT**
- B) They are the same thing
- C) Vulnerability = attacker, Threat = defense

Explanation: Threat = what CAN happen (hacker attack). Vulnerability = weakness that makes it possible (outdated software).

🎯 Scoring System

- Each correct answer: +50 points
- Total possible: 200 points
- Score displayed: `Score: X/200`

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2. Network Basics

📋 Overview

- **Tutorial ID:** `beginner_network`
- **Duration:** ~10 minutes
- **Format:** Multi-section learning + Quiz
- **Max Score:** 150 points (3 questions × 50 points each)
- **XP Rewards:** Same as Fundamentals

📚 Learning Sections

Section 1: Introduction

Understanding how computers communicate:

- 🌐 IP Addresses (computer addresses)
- 🚪 Ports (doors for services)
- 📡 Protocols (languages computers speak)

Section 2: IP Addresses

Definition: Unique identifier for computers on a network.

Two Types:

1️⃣ PRIVATE IP (Local Network Only)

- 192.168.x.x, 10.x.x.x, 172.16-31.x.x
- Only visible on your home/office network
- Router assigns these

- Example: Your laptop = 192.168.1.100

2 PUBLIC IP (Internet-Facing)

- Visible to entire internet
- ISP assigns one per router
- Example: Your home router = 45.67.89.123
- All devices share this PUBLIC IP

Network Diagram:

```
[Your Laptop]          [Your Phone]
192.168.1.100          192.168.1.101
       ↓                      ↓
    [Home Router] ← Public IP: 45.67.89.123
           ↓
      [Internet]
           ↓
  [Attacker Server] ← Public IP: 45.33.32.156
```

Section 3: Ports

Definition: Apartment numbers for services on one computer.

Format: `IP:Port` → `192.168.1.100:80`

Common Ports:

- ✓ Port 80 = HTTP (websites)
- ✓ Port 443 = HTTPS (secure websites)
- ✓ Port 22 = SSH (remote login)
- ✓ Port 25 = SMTP (email)
- ✓ Port 3389 = RDP (Windows Remote Desktop)

⚠ **Suspicious Ports (malware):**

- ✕ Port 4444 = Common backdoor
- ✕ Port 31337 = "Elite" hacker port
- ✕ Port 1337 = Hacker favorite

Example: `Connection to 45.33.32.156:4444` → 🚨 RED FLAG!

Section 4: Protocols

Definition: Rules for how computers communicate.

Web Protocols:

- **HTTP** = Not encrypted (anyone can read data)

- **HTTPS** = Encrypted (safe for passwords)

Email Protocols:

- **SMTP** = Send email (port 25)
- **POP3/IMAP** = Receive email (ports 110/143)

Network Protocols:

- **TCP** = Reliable delivery (confirms packets)
- **UDP** = Fast but unreliable (streaming, gaming)

Security Protocols:

- **SSH** = Secure remote login (port 22)
- **TLS/SSL** = Encryption layer (used in HTTPS)

Protocol Stack:

```
Application:  HTTP, FTP, SSH
     ↓
Transport:    TCP, UDP
     ↓
Internet:     IP (routing)
     ↓
Physical:     WiFi, Ethernet
```

✅ Quiz Questions & Answers

Question 1

Q: You see traffic to 45.33.32.156:4444. What does port 4444 indicate?

Options:

- A) Normal web browsing
- B) Suspicious backdoor port ✅ **CORRECT**
- C) Email connection

Explanation: Port 4444 is commonly used by backdoor trojans! Port 80/443 = web, Port 4444 = suspicious!

Question 2

Q: What's the difference between public and private IP addresses?

Options:

- A) Public = internet, Private = local network only ✅ **CORRECT**
- B) They are the same

- C) Private = faster, Public = slower

Explanation: Private IPs (192.168.x.x) work only on local network. Public IPs are visible to the entire internet!

Question 3

Q: Which protocol is secure (encrypted)?

Options:

- A) HTTP
- B) HTTPS ✅ **CORRECT**
- C) FTP

Explanation: HTTPS uses encryption (the 'S' = Secure). HTTP sends data in plaintext - anyone can read it!

🎯 Scoring System

- Each correct answer: +50 points
- Total possible: 150 points

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3. Encryption Basics

📋 Overview

- **Tutorial ID:** `beginner_encryption`
- **Duration:** ~12 minutes
- **Format:** Demo + Challenge + Education
- **Max Score:** 100 points
- **XP Rewards:** Same as other beginner tutorials
- **Minimum Score:** 50 points (after 3 wrong attempts)

📚 Learning Phases

Phase 1: Introduction

What is Encryption?

Encryption transforms readable text (plaintext) into unreadable code (ciphertext).

Example:

```
Original: "HELLO"
Encrypted: "KHOOR"
```

Only someone with the KEY can decrypt it back.

Phase 2: Encryption Demo

Caesar Cipher Demo

Each letter shifts by a number (the KEY).

Try it:

1. Enter a message: `HELLO WORLD`
2. Set shift key: `3`
3. Click ENCRYPT
4. Result: `KHOOR ZRUOG`

Phase 3: Decryption Challenge

Your Turn!

Encrypted Message: `WKLV LV VHFUHW`
Key: 3

Task: Decrypt by shifting each letter BACKWARDS by 3.

Solution:

```
W → T
K → H
L → I
V → S

Answer: "THIS IS SECRET"
```

Scoring:

- Correct on 1st try: +100 points
- Each wrong attempt: -10 points
- After 3 wrong attempts: Auto-show answer, 50 points minimum

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Phase 4: Ransomware Explanation

Why Ransomware is Dangerous

Caesar Cipher: Easy to crack (only 26 keys)

Modern Ransomware: AES-256 encryption

- 2^256 possible keys (more than atoms in universe!)
- Billions of years to try all keys
- Mathematically impossible to crack

How Ransomware Works:

1. Uses AES-256 (unbreakable encryption)
2. Deletes key from your computer
3. Attacker keeps only copy of key
4. Demands payment for key

Your Defense:

- ✅ Daily backups
- ✅ Air-gapped backups (offline)
- ✅ Cloud + local copies
- ❌ NEVER pay ransom

🎯 Scoring System

- Correct answer: 100 points
- Wrong attempts: -10 points each
- Minimum guarantee: 50 points (after 3 attempts)

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4. Malware Types Identification

📋 Overview

- **Tutorial ID:** `beginner_malware`
- **Duration:** 90 seconds (timed!)
- **Format:** Drag-and-drop identification game
- **Max Score:** 100 points
- **Total Incidents:** 6
- **XP Rewards:** Same as other tutorials

🎮 How to Play

1. **Read incident reports** - Each shows symptoms
2. **Drag malware buttons** - Drag correct type to report
3. **Beat the clock** - 90 seconds to identify all 6
4. **Score breakdown:**
   - Completion: 50 points (max)
   - Time bonus: 50 points (max)
   - Formula: `score = (reports_solved/6 × 50) + (time_remaining/90 × 50)`

📋 Incident Reports & Answers

Incident #2847

Symptoms:

- All files encrypted with .locked extension

- Ransom note demanding Bitcoin payment
- Desktop wallpaper changed to payment instructions

Answer: 🔒 **RANSOMWARE**

Explanation: Ransomware encrypts files and demands payment in cryptocurrency

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Incident #1923

Symptoms:

- Pop-up advertisements appearing constantly
- Browser homepage changed without permission
- System running slower than usual

Answer: 📢 **ADWARE**

Explanation: Adware displays unwanted ads and tracks browsing habits

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Incident #5634

Symptoms:

- Program disguised as video game installer
- Creates backdoor access for remote control
- Firewall alerts about unauthorized connections

Answer: 🐴 **TROJAN**

Explanation: Trojans disguise themselves as legitimate software

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Incident #7721

Symptoms:

- Keylogger recording all passwords
- Banking credentials stolen
- Runs silently in background processes

Answer: 🕵️ **SPYWARE**

Explanation: Spyware secretly monitors activity and steals personal data

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Incident #3309

Symptoms:

- Infected email attachment executed
- Malware attaches to .exe and .doc files

- Spreads when files are shared via USB

Answer: 🦠 **VIRUS**

Explanation: Viruses self-replicate by attaching to files

Incident #8156

Symptoms:

- Spreads automatically across entire network
- Does not require user interaction
- Consumes massive bandwidth

Answer: 🐛 **WORM**

Explanation: Worms self-replicate and spread across networks automatically

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📚 Reference Sheet

Available during game:

🦠 VIRUS

- Attaches to files
- Spreads via USB, email
- Requires user to run file

🐛 WORM

- Spreads automatically
- No user interaction needed
- Consumes network bandwidth

🐴 TROJAN

- Disguises as legitimate software
- Creates backdoors
- Remote access tool

🔒 RANSOMWARE

- Encrypts files
- Demands payment
- Bitcoin/cryptocurrency

🕵️ SPYWARE

- Monitors activity
- Steals passwords
- Keyloggers, screen capture

📢 ADWARE

- Shows ads
- Tracks browsing
- Slows system

🎯 Scoring System

```
Completion Score = (Reports Solved / 6) × 50
Time Bonus = (Time Remaining / 90) × 50
Final Score = Completion Score + Time Bonus
```

Example:

- Solved 6/6 reports
- Time remaining: 45 seconds
- Completion: 50 points
- Time bonus: 25 points
- **Final: 75/100**

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5. Phishing Lab

📋 Overview

- **Tutorial ID:** `beginner_phishing`
- **Duration:** 90 seconds (timed!)
- **Format:** Email classification game
- **Total Emails:** 8
- **Max Score:** 1200 points (8 emails × 150 points each)
- **XP Rewards:** Based on percentage (90%+ = 200 XP)

🎮 How to Play

1. **Read each email** carefully
2. **Click SAFE or PHISHING** button
3. **Learn from feedback** - Red flags or legitimate signs
4. **Beat the clock** - 90 seconds total

Scoring:

- Correct identification: +150 points
- Wrong identification: -50 points

📧 Email 1: PayPal Phishing

From: security@paypa1-secure.tk
Subject: 🚨 URGENT: Verify Your Account Now!

Body:

```
Dear Customer,

Your PayPal account will be SUSPENDED in 24 hours
if you don't verify your identity immediately!

Click here to verify: http://paypal-verify.tk/login

Best regards,
PayPal Security Team
```

Answer: 🚨 PHISHING

Red Flags:

- Generic greeting ('Dear Customer')
- Urgent threatening language
- Fake domain (.tk instead of .com)
- Typo in sender: 'paypa1' not 'paypal'
- Suspicious link domain

📨 Email 2: Legitimate Netflix Receipt

From: noreply@netflix.com
Subject: Your Netflix receipt

Body:

```
Hi Sarah,

Your $15.99 payment for Netflix Premium was processed
successfully on November 29, 2025.

Watch history this month:
- Stranger Things S4
- The Crown S6

Thank you for being a member!

The Netflix Team
```

Answer: ✅ SAFE (Legitimate)

Legitimate Signs:

- Personal greeting (uses your name)
- No urgent action required
- Official @netflix.com domain
- Specific transaction details
- No suspicious links

📨 Email 3: Password Reset Phishing

From: it-support@company-email.com
Subject: Password Expiration Warning

Body:

```
Your company password will expire tomorrow.

Please click this link to update your password:
http://bit.ly/pwdreset2025

IT Support
```

Answer: 🚨 PHISHING

Red Flags:

- Suspicious shortened URL (bit.ly)
- Generic 'IT Support' signature
- Unexpected password reset request
- No company branding or logo
- Creates false urgency

📨 Email 4: Legitimate Amazon Order

From: support@amazon.com
Subject: Your Amazon.com order #402-1849302-7829103

Body:

```
Hello John,

Your order has been shipped!

Order Details:
- Wireless Mouse (Black) - $24.99
- Delivery: Dec 2, 2025
- Track package: [View in Amazon account]

Amazon Customer Service
```

Answer: ✅ **SAFE (Legitimate)**

Legitimate Signs:

- Real Amazon order number format
- Personal name used
- Specific product details
- No external links (directs to account)
- Official @amazon.com domain

📨 Email 5: Lottery Scam

From: prize-winner@lottery-international.org
Subject: 🎉 YOU'VE WON $1,000,000! Claim Now!

Body:

```
CONGRATULATIONS!

You have been selected as the WINNER of our
International Lottery!

Prize: $1,000,000 USD

To claim your prize, send your:
- Full name
- Address
- Bank account number
- Social security number

Reply within 48 hours or forfeit!
```

Answer: 🚨 **PHISHING**

Red Flags:

- 'Too good to be true' prize
- Requests sensitive information
- Suspicious sender domain (.org)
- High-pressure deadline
- You never entered a lottery!
- Requests bank/SSN info via email

📨 Email 6: Legitimate GitHub Security

From: notifications@github.com
Subject: [GitHub] Password changed successfully

Body:

```
Hi @username,

Your GitHub password was changed on November 29, 2025
at 10:45 AM PST.

If you didn't make this change, please secure your
account immediately:
https://github.com/settings/security

GitHub Security Team
```

Answer: ✅ SAFE (Legitimate)

Legitimate Signs:

- Official @github.com domain
- Specific date/time of action
- Includes your GitHub username
- Legitimate security link
- Standard security notification

📨 Email 7: CEO Fraud (Business Email Compromise)

From: ceo@company.com
Subject: URGENT: Wire Transfer Needed

Body:

```
I'm in a meeting and need you to process an urgent
wire transfer immediately.

Amount: $50,000
Account: [Bank details attached]

This is confidential. Don't discuss with anyone.

- CEO
```

Answer: 🚨 PHISHING

Red Flags:

- CEO Fraud / Business Email Compromise
- Creates false urgency
- Requests large money transfer

- Orders to keep it secret
- Unusual request from executive
- Should verify via phone call!

📨 Email 8: Legitimate Slack Invite

From: team@slack.com
Subject: You're invited to join Engineering Team workspace

Body:

```
Hi Alex,

John Doe invited you to join the 'Engineering Team'
workspace on Slack.

Click to join: https://slack.com/accept-invite/T01234/B56789

Workspace: Engineering Team (acme-corp.slack.com)

The Slack Team
```

Answer: ✅ SAFE (Legitimate)

Legitimate Signs:

- Official @slack.com domain
- Specific workspace name shown
- Legitimate Slack invite URL structure
- Names who invited you
- Standard Slack invite format

🎯 Scoring System

```
Correct: +150 points
Wrong: -50 points
Max Score: 1200 points (8 × 150)

Grade Scale:
90%+: Grade A
80-89%: Grade B
70-79%: Grade C
60-69%: Grade D
<60%: Grade F
```

🟡 INTERMEDIATE TUTORIALS

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6. Hands-On Malware Removal Lab

📋 Overview

- **Tutorial ID:** `intermediate_defense`
- **Duration:** ~15-20 minutes
- **Format:** Interactive command-line simulation
- **Max Score:** 1000 points
- **Phases:** 5 (Download → Infection → Detection → Containment → Eradication)

🎮 How to Play

You are the victim! Experience a malware infection firsthand, then use CMD commands to remove it.

Phase Flow:

1. **Download:** Click fake game installer
2. **Infection:** Malware executes
3. **Detection:** Use CMD to find malware
4. **Containment:** Kill processes & block C2
5. **Eradication:** Remove all traces

📚 Available Commands

Detection Commands

`tasklist` - List all running processes

```
Output:
explorer.exe           PID: 1234    Normal
svchost.exe           PID: 5678    Normal
malware.exe           PID: 9012    ⚠️ SUSPICIOUS
svchost32.exe         PID: 3456    ⚠️ SUSPICIOUS (Fake svchost)
updater.exe           PID: 7890    ⚠️ SUSPICIOUS
```

`netstat -ano` - Show network connections

```
Output:
192.168.1.100 → 8.8.8.8:443          [HTTPS - Normal]
192.168.1.100 → 45.33.32.156:443     ⚠️ [C2 SERVER - MALWARE]
```

`dir %appdata%` - Check AppData folder

```
Output:
malware.exe           120 KB    ⚠ MALICIOUS
svchost32.exe          95 KB    ⚠ MALICIOUS
```

Containment Commands

`taskkill /F /IM malware.exe` - Kill malicious process

```
Result: ✅ Process terminated successfully
```

`taskkill /F /IM svchost32.exe` - Kill fake svchost

```
Result: ✅ Process terminated successfully
```

`taskkill /F /IM updater.exe` - Kill updater process

```
Result: ✅ Process terminated successfully
```

`netsh advfirewall firewall add rule name="blockmfw" dir=out remoteip=45.33.32.156 action=block`

```
Result: ✅ Firewall rule added - C2 server blocked!
```

Eradication Commands

`reg delete "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v "WindowsUpdate" /f`

```
Result: ✅ Registry key deleted - Persistence removed!
```

`del /F /Q "%appdata%\malware.exe"`

```
Result: ✅ Malicious file deleted!
```

`del /F /Q "%appdata%\svchost32.exe"`

```
Result: ✅ Malicious file deleted!
```

`schtasks /delete /tn "SystemUpdate" /f`

```
Result: ✅ Scheduled task removed!
```

🎯 Complete Solution (Step-by-Step)

Phase 3: Detection

```
tasklist
netstat -ano
```

Goal: Identify malicious processes and C2 connection
Completion Bonus: +100 points

Phase 4: Containment

```
taskkill /F /IM malware.exe
taskkill /F /IM svchost32.exe
taskkill /F /IM updater.exe
netsh advfirewall firewall add rule name="blockmfw" dir=out remoteip=45.33.32.156
action=block
```

Goal: Stop malware execution and block C2
Completion Bonus: +100 points

Phase 5: Eradication

```
reg delete "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v "WindowsUpdate"
/f
del /F /Q "%appdata%\malware.exe"
del /F /Q "%appdata%\svchost32.exe"
schtasks /delete /tn "SystemUpdate" /f
```

Goal: Remove all malware traces
Completion Bonus: +150 points

🎯 Scoring System

```
Starting Score: 1000 points
Wrong commands: -10 points each
Detection completion: +100 points
Containment completion: +100 points
Eradication completion: +150 points
Time bonus: +0 to +180 points (under 3 minutes)

Grade Scale:
900+: S-RANK (Cyber Expert!)
750-899: A-RANK (Security Pro!)
600-749: B-RANK (Good Work!)
<600: C-RANK (Keep Learning!)
```

💡 Hints System

- Hints appear after 15 seconds of inactivity
- No penalty for using hints
- Phase-specific guidance

Detection Phase Hints:

- "Type 'tasklist' to see running processes"
- "Type 'netstat -ano' to check network connections"

Containment Phase Hints:

- "Use 'taskkill /F /IM malware.exe' to stop the process"

Eradication Phase Hints:

- "Use 'reg delete' to remove persistence keys"
- "Use 'del /F /Q' to delete malicious files"

🥷 ADVANCED TUTORIALS

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7. Malware Incident Response Simulation

📋 Overview

- **Tutorial ID:** `advance_scenarios`
- **Duration:** ~10-15 minutes
- **Format:** Multiple-choice decision making
- **Max Score:** 1000 points
- **Phases:** 5 (Detection → Investigation → Containment → Eradication → Recovery)

🎮 How to Play

Realistic incident response simulation. Make decisions at each phase of a malware attack.

Scoring:

- Start: 1000 points
- Correct action: +50 bonus
- Wrong action: -150 to -200 penalty

📋 Scenario 1: DETECTION

Alert:

```
SECURITY ALERT: Employee John Doe clicked 'invoice.pdf.exe'
Endpoint Detection: Suspicious process spawned on PC-ACCOUNTING-07
```

Your Actions:

- A) Delete the file immediately
- B) Isolate machine from network ✅ **CORRECT**
- C) Restart the computer

Correct Answer: B) Isolate machine from network

Why?

- ✅ Network isolation prevents lateral movement
- ✅ Stops C2 communication
- ❌ Deleting/restarting triggers fail-safe mechanisms
- ❌ Malware may spread or encrypt immediately

Penalty for Wrong: -200 points

📋 Scenario 2: INVESTIGATION

Alert:

```
Machine isolated. Analyzing system state...
What should you investigate FIRST?
```

Your Actions:

- A) Check browser history
- B) Analyze network connections & processes ✅ **CORRECT**
- C) Scan with antivirus

Correct Answer: B) Analyze network connections & processes

Why?

- ✅ Network connections reveal C2 servers
- ✅ Shows what attacker is doing RIGHT NOW
- ❌ Browser history too slow
- ❌ AV scan takes too long

Discovery:

```
You found:
- C2 Server: 45.33.32.156:443
- Process: invoice.exe spawned hidden PowerShell scripts
```

Penalty for Wrong: -150 points

📋 Scenario 3: CONTAINMENT

Alert:

```
Threat identified: Trojan with C2 connection to 45.33.32.156
What is your NEXT containment action?
```

Your Actions:

- A) Unplug the network cable only
- B) Kill processes + Block C2 IP at firewall ✅ **CORRECT**
- C) Shutdown the computer

Correct Answer: B) Kill processes + Block C2 IP at firewall

Why?

- ✅ Multi-layer containment is best
- ✅ Killing processes stops execution
- ✅ Firewall block prevents reconnection
- ❌ Unplugging alone: Malware persists, reconnects after reboot
- ❌ Shutdown: Loses volatile memory evidence (RAM)

Penalty for Wrong: -150 points

📋 Scenario 4: ERADICATION

Alert:

```
Processes killed, C2 blocked. Removing persistence mechanisms...
Where does this Trojan typically hide?
```

Your Actions:

- A) Only in Downloads folder
- B) Registry Run keys + Scheduled Tasks + AppData ✅ **CORRECT**
- C) Desktop shortcuts

Correct Answer: B) Registry Run keys + Scheduled Tasks + AppData

Why?

- ✅ Trojans use multiple persistence methods
- ❌ If you miss registry keys or scheduled tasks, malware auto-restarts

Removed:

```
✅ HKCU\...\Run\WindowsUpdate
✅ Scheduled Task: 'SystemUpdate'
✅ C:\Users\JohnDoe\AppData\Roaming\svchost32.exe
```

Penalty for Wrong: -150 points

📋 Scenario 5: RECOVERY

Alert:

```
Malware removed successfully. Final recovery step?
How do you ensure system integrity?
```

Your Actions:

- A) Just restart and continue working
- B) Restore from clean backup + Patch vulnerability + Train user ✅ **CORRECT**
- C) Reinstall OS (nuclear option)

Correct Answer: B) Restore from backup + Patch + Train user

Why?

- ✅ Defense-in-depth recovery strategy
- ✅ Clean backup ensures no remnants
- ✅ Patch closes attack vector
- ✅ User training prevents re-infection
- ❌ Restarting alone: Rootkit components may remain

- ❌ Full reinstall: Overkill, causes downtime

Result: 🎯 INCIDENT RESOLVED!

Penalty for Wrong: -150 points

🎯 Scoring System

```
Starting Score: 1000 points
Correct answers: +50 bonus each
Wrong answers: -150 to -200 penalty
Time bonus: +0 to +300 (under 5 minutes)

Final Grade:
900+: S-RANK (Security Expert! 🏆)
750-899: A-RANK (Incident Response Pro! 🥇)
600-749: B-RANK (Good Security Awareness 🥈)
400-599: C-RANK (Needs More Training 🥉)
<400: D-RANK (Review the Basics 📚)
```

---

8. Trojan Horse Tutorial (Hacker POV)

📋 Overview

- **Tutorial ID:** Special educational tutorial
- **Duration:** ~10 minutes
- **Format:** Narrative walkthrough from attacker perspective
- **Purpose:** Educational - understand attack to better defend
- **Phases:** 6 (Create → Disguise → Deploy → Execute → Backdoor → Defense)

⚠️ Disclaimer

This is EDUCATIONAL ONLY! Understanding how attacks work helps you defend against them.

📚 Phase-by-Phase Breakdown

Phase 1: Create Payload

Objective: Create malicious backdoor trojan

Code Snippet:

```
// Trojan code snippet:
function createBackdoor() {
    openPort(4444);
    connectToC2("attacker.com");
```

```
    hideProcess();
    setPersistence();
}
```

What It Does:

- Opens port 4444 for remote access
- Connects to Command & Control server
- Hides from Task Manager
- Auto-starts on boot

Defense Tip: 🛡️
Always scan downloaded files with antivirus before running!

Phase 2: Disguise

Objective: Make malware look legitimate

Disguise Settings:

```
filename = "SuperGame_Setup.exe"
icon = "game_icon.ico"
description = "Official Installer"
publisher = "GameStudio Inc."
certificate = FORGED
```

Techniques:

- Use game/software name
- Add realistic icons
- Fake publisher information
- Forged digital signature

Defense Tip: 🛡️
Check publisher certificates! Verify software is from official sources.

Phase 3: Deploy

Objective: Upload trojan to distribution sites

Deployment Targets:

```
✓ Free-Games-Download.com
✓ TorrentSite.org
✓ SoftwarePortal.net
✓ Email attachment (phishing)
```

```
Status: UPLOADED
Views: 1,247 downloads
```

Distribution Methods:

- File-sharing sites
- Torrent networks
- Fake download portals
- Phishing emails

Defense Tip: 🛡️
Only download from official websites. Avoid torrents and file-sharing!

Phase 4: Execute

Objective: Victim runs the trojan

Execution Log:

```
[+] Trojan executed successfully
[+] Admin privileges: NO (using UAC bypass)
[+] Disabling Windows Defender...
[+] Creating persistence...
[+] Establishing connection...
```

What Happens:

- Trojan runs with user privileges
- Bypasses UAC (User Account Control)
- Disables antivirus
- Creates registry entries
- Connects to C2 server

Victim's Experience:

```
"Weird... the installer crashed.
Maybe I'll try again later..."
```

Defense Tip: 🛡️
Never disable antivirus to run installers. That's a red flag!

Phase 5: Backdoor Access

Objective: Attacker gains full remote access

Backdoor Active:

```
C2 Server: CONNECTED
Port 4444: OPEN
Privileges: USER

Available commands:
> download_files
> capture_screen
> log_keystrokes
> steal_passwords
```

Attacker Capabilities:

- Access file system
- Control webcam
- Log keystrokes
- Steal passwords
- Install more malware

Victim Status:

```
🔴 COMPROMISED

"Everything seems normal..."

[Hidden processes running]
[Data being exfiltrated]
```

Phase 6: Defense Lesson

🛡️ HOW TO DEFEND AGAINST TROJANS

Prevention:

- ✅ Only download from official websites
- ✅ Verify digital signatures
- ✅ Keep antivirus updated
- ✅ Enable firewall
- ✅ Don't disable security software

Detection:

- ✅ Monitor network connections (netstat)
- ✅ Check running processes (Task Manager)
- ✅ Look for suspicious ports (4444, 31337)
- ✅ Use anti-malware scanners

- ✅ Watch for unusual firewall alerts

Removal:

- ✅ Kill malicious processes
- ✅ Block C2 IP at firewall
- ✅ Remove registry persistence
- ✅ Delete malicious files
- ✅ Restore from clean backup

Red Flags:

- ❌ Software from unofficial sources
- ❌ Installer asks to disable antivirus
- ❌ Unexpected network connections
- ❌ Hidden processes in AppData
- ❌ Firewall alerts to port 4444

🎯 Key Takeaways

What You Learned:

1. How trojans are created and disguised
2. Distribution methods attackers use
3. How trojans gain persistence
4. What backdoor access looks like
5. How to defend your system

Remember:

- Understanding attacks = Better defense
- Never download from untrusted sources
- Verify software authenticity
- Keep security software enabled
- Monitor your system regularly

📊 XP & Progression System

XP Rewards by Tutorial

Tutorial	Max Score	100% XP	75%+ XP	50-74% XP	<50% XP
Beginner Fundamentals	200	200	150	100	0
Beginner Network	150	200	150	100	0
Beginner Encryption	100	200	150	100	0
Beginner Malware	100	200	150	100	0

Tutorial	Max Score	100% XP	75%+ XP	50-74% XP	<50% XP
Beginner Phishing	1200	200	150	100	0
Intermediate Defense	1000	200	150	100	0
Advanced Scenarios	1000	200	150	100	0

Rank System

Rank	XP Required	Icon
Iron	0 - 499	🔩
Bronze	500 - 999	🥉
Silver	1000 - 1499	🥈
Gold	1500 - 1999	🥇
Platinum	2000 - 2499	💎
Diamond	2500 - 2999	💠
Master	3000 - 3499	⭐
Challenger	3500+	👑

🎮 Tutorial Navigation Flow

```
Landing Page
    ↓
Mode Selection
    ↓
┌─────────────────────────────────────────┐
│  BEGINNER TUTORIALS                     │
│  ├─ Cybersecurity Fundamentals         │
│  ├─ Network Basics                     │
│  ├─ Encryption Basics                  │
│  ├─ Malware Types                      │
│  └─ Phishing Lab                       │
└─────────────────────────────────────────┘
    ↓
┌─────────────────────────────────────────┐
│  INTERMEDIATE TUTORIALS                 │
│  └─ Hands-On Malware Removal Lab       │
└─────────────────────────────────────────┘
    ↓
┌─────────────────────────────────────────┐
│  ADVANCED TUTORIALS                     │
│  ├─ Malware Incident Response          │
│  ├─ Trojan Tutorial (Hacker POV)       │
```

```
│   └─ [More coming soon]                │
└───────────────────────────────────────┘
    ↓
Landing Page (XP & Progress Saved)
```

📝 Quick Reference Sheets

Command Cheat Sheet (Intermediate Lab)

Detection

```
tasklist                # List processes
netstat -ano            # Network connections
dir %appdata%           # Check AppData folder
help                    # Show command help
```

Containment

```
taskkill /F /IM [process.exe]      # Kill process
netsh advfirewall firewall add rule name="block" dir=out remoteip=[IP]
action=block
```

Eradication

```
reg delete "HKCU\...\Run" /v "[KeyName]" /f     # Remove registry
del /F /Q "[filepath]"                          # Delete file
schtasks /delete /tn "[TaskName]" /f            # Remove task
```

Malware Types Quick Reference

Type	How It Spreads	Main Goal	Example
Virus	Files (USB, email)	Replicate, damage	CIH, ILOVEYOU
Worm	Network (automatic)	Spread rapidly	WannaCry, Conficker
Trojan	Disguise as software	Backdoor access	Zeus, Emotet
Ransomware	Phishing, exploits	Encrypt files	Locky, Ryuk
Spyware	Downloads, bundles	Steal data	Keyloggers
Adware	Free software	Show ads	Browser hijackers

Phishing Red Flags Checklist

- ❌ Generic greeting ("Dear Customer")
- ❌ Urgent/threatening language
- ❌ Suspicious sender domain
- ❌ Typos in sender address
- ❌ Shortened URLs (bit.ly)
- ❌ Requests for passwords/SSN
- ❌ Too good to be true offers
- ❌ High-pressure deadlines
- ❌ Unexpected attachments
- ❌ Poor grammar/spelling

🏆 Achievement System

Tutorial Completion Badges

Badge	Requirement	Reward
🎓 **Fundamentals Master**	Complete CIA Triad tutorial	200 XP
🌐 **Network Ninja**	Complete Network Basics	200 XP
🔒 **Crypto Expert**	Complete Encryption tutorial	200 XP
🦠 **Malware Hunter**	Complete Malware Types	200 XP
🕵️ **Phishing Detective**	Complete Phishing Lab	200 XP
⚔️ **Defender**	Complete Malware Removal Lab	200 XP
🚨 **Incident Responder**	Complete Advanced Scenario	200 XP

Special Achievements

Badge	Requirement	Reward
🥇 **Perfect Score**	Get 100% on any tutorial	Bonus 50 XP
⚡ **Speed Demon**	Complete timed tutorial in half time	Bonus 100 XP
🎯 **All Cleared**	Complete all beginner tutorials	Bonus 200 XP
👑 **Elite Security**	Reach Challenger rank	Special badge

📖 Glossary

C2 (Command & Control): Server that attackers use to control malware remotely.

CIA Triad: Confidentiality, Integrity, Availability - core security principles.

Encryption: Converting data into unreadable code using a key.

Firewall: Security system that monitors and controls network traffic.

Payload: Malicious code that performs the main attack function.

Persistence: Technique malware uses to survive reboots.

Phishing: Social engineering attack using fake emails/websites.

Registry: Windows database storing system/application settings.

UAC (User Account Control): Windows security feature requiring permission for admin actions.

Zero-Day: Vulnerability unknown to software vendor, no patch available.

📞 Support & Resources

Having Trouble?

1. **Check hints** - Wait 15 seconds for automatic hints
2. **Use help command** - Type `help` in CMD tutorials
3. **Review tutorial sections** - Use BACK button to revisit
4. **Reference sheets** - Built-in guides available

Report Issues

If you encounter bugs or have suggestions:

- File an issue on the GitHub repository
- Contact game support team

Document Version: 1.0
Last Updated: December 3, 2025
Total Tutorials: 8 (5 Beginner, 1 Intermediate, 2 Advanced)
Total Questions: 75+ with complete answers

🎉 Good Luck, Security Trainee!

Remember: The best defense is knowledge. Complete all tutorials to become a cybersecurity expert!

Pro Tips:

- ✅ Read all feedback carefully
- ✅ Take your time on quizzes
- ✅ Practice command-line labs multiple times
- ✅ Apply what you learn to real-world security

- ✅ Always verify before clicking links!

Stay Safe Online! 🔒